

A Compositional, Audit-Ready Proof Program for 3D Navier–Stokes

Abstract

We present a windowed, auditable proof program for dissipative PDEs. Analytical risk lives in a graded algebra $\mathcal{Q}$; side-conditions in a typed effect system Σ ; and a lax composition rule `compose_ev` upper-bounds order effects. A Hybrid-Space layer prices reassociation (time $\leftrightarrow$ frequency $\leftrightarrow$ patch $\leftrightarrow$ mesh) as frontier crossings with No-Zeno dwell, Dirac freeze/reproject, and a geometry-priced frontier defect that prints into $\mathcal{Q}$. Specializing to 3D incompressible Navier–Stokes (NSE), the spine reduces to five lemmas: (L1) a frequency-local energy inequality under a frozen Littlewood–Paley (LP) architecture, (L2) a priced-commutator interchange bound, (L3) ε -regularity via a CVaR $\rightarrow$ CKN bridge, (L4) closure in $L_t^\infty L_x^3$, and (L5) a rigidity theorem for ancient limits. Discharging L4 empties Σ at the first candidate singular time; the solution extends.

1 Introduction and contributions

Problem context. The global regularity of 3D incompressible Navier–Stokes (NSE) is open. Classical strategies struggle to control intermittent, cascade-prone dynamics in a way that remains auditably correct under operational changes (basis swaps, mesh retunes, windowing). Our approach builds a *proof program*: correctness is certified by deterministic replay against a public acceptance surface rather than by narrative-only prose.

Audit-ready paradigm. Evidence is ingested as a Raw monoid and interpreted by monotone-homomorphism adapters into a graded acceptance algebra $\mathcal{Q}$ with max- and sum-channels aggregated by $\wedge$. A typed effect system Σ tracks obligations, and a lax composition rule `compose_ev` bounds order effects. A three-step $V1 \rightarrow V2 \rightarrow V3$ pipeline enforces schema, formation, and aggregation discipline, while a determinism envelope (hash suite, container hash, seeded clocks, tolerances) renders every gate replayable. Throughout we adopt the convention *dimensionless unless stated* and provide units where relevant.

Two core innovations. (I) *Priced frontiers.* Non-commuting reassociations are modeled as *frontier crossings* that incur a *saltation bound* C_{salt} and a ledgered Δ_{fr} . We explicitly enforce units: dF/ν is dimensionless; so are C_{salt} , K_{obs} , and the frontier sum-channel Q_{fr} . By design, Δ_{fr} debits *only* sum-channels, preserving max-channel invariants used by decisive L^∞ -type gates. (II) *CVaR $\rightarrow$ CKN bridge.* Tail-risk control on the scaled local energy (CVaR $_{1-\alpha}$) deterministically yields a $(1 - \alpha)$ -mass subcover satisfying a CKN-type smallness criterion. The bridge constant is digest-bound: $C_{\text{CKN}} = C_0 N_V C_{\text{CZ}} C_{\text{LEI}}$.

Roadmap. Part I formalizes the calculus and acceptance surfaces. Part II develops the NSE lemmas with auditable constants. Appendices supply harnesses and fixtures, and outline an empirical validation protocol.

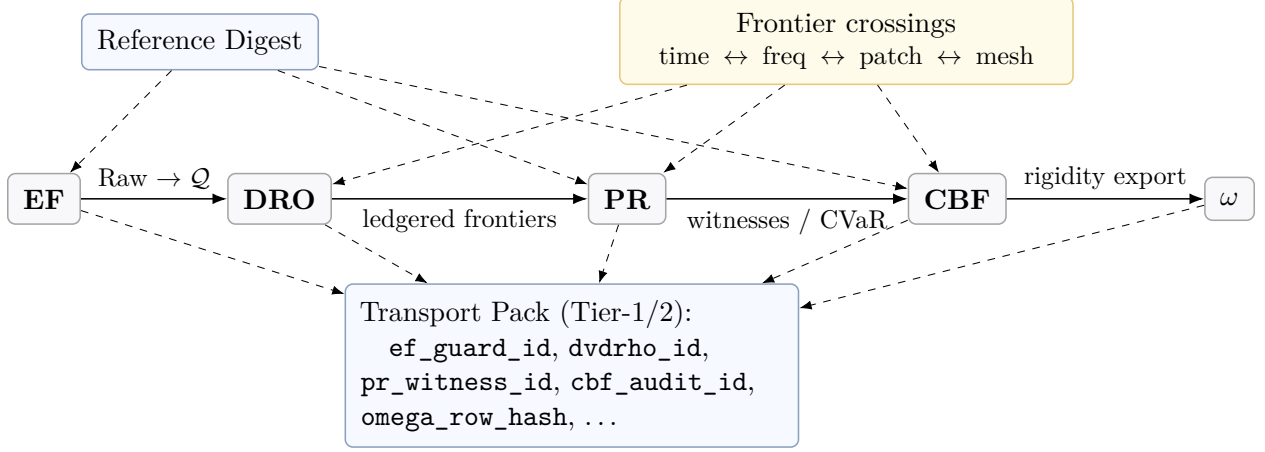


Figure 1: End-to-end audit pipeline: stages read/write the Transport Pack and consult the Reference Digest; frontier crossings are priced and ledgered.

Contributions (paper structure).

1. **Part I (Problem-agnostic calculus).** Defines $\mathcal{Q}$, Σ , compose_ev ; proves the Laxity Lemma under explicit preconditions; formalizes I1–I5 including the unknowns rule and hash/alias checks; fixes units for frontier spend. *Acceptance map:* I1, I2, I4a, I4b, I5.
2. **Part II (NSE spine L1–L5).** Pins constants for L1 (frozen LP + CZ split), states the L4 two-block inequality, and exports rigidity hypotheses at ω . *Acceptance map:* L1→I1, L3→I4a, L4→L3GATE, ω →export.
3. **Audit surface & artifacts.** Canonical `proof_print`, Transport Pack schema (with alias rule), replay harnesses, and negative controls.

2 Part I — Problem-agnostic calculus and audit surface ($\mathcal{Q}$, Σ , compose_ev)

Standing constants (Digest bindings).

$\text{cz_kernel_id} \mapsto C_{\text{CZ}}$, $\text{lei_norm_const} \mapsto C_{\text{LEI}}$, $\text{nv_multiplicity} \mapsto N_V$, $\text{dvdrho_id} \mapsto \tau_{\min}$, $\text{cbf_audit_id} \mapsto (c_{\Sigma}, \kappa_{\max}, r_{\text{tube}}, \hat{\nu}_{\min})$.

2.1 Acceptance algebra $\mathcal{Q}$: channels, units, unknowns, Raw monoid

Carrier, orders, and channels. Grades are vectors $q \in \mathcal{Q} = \prod_i \mathcal{Q}_i$ with coordinatewise preorder (*larger=worse*). We fix disjoint index sets of channels:

$$\mathcal{S} = \{L, \mathcal{E}_{\text{CVaR}}, Q_{\text{fr}}, \dots\}, \quad \mathcal{M} = \{E, S_3, P_{H^{-1}}, Q_{\tau}, Q_h, \dots\}.$$

Aggregation $\wedge$ acts *coordinatewise* by

$$(q \wedge q')_i = \begin{cases} \max\{q_i, q'_i\}, & i \in \mathcal{M}, \\ q_i + q'_i, & i \in \mathcal{S}. \end{cases}$$

We write $\mathcal{Q}_{\text{sum}}$ and $\mathcal{Q}_{\text{max}}$ for the subvectors on $\mathcal{S}$ and $\mathcal{M}$.

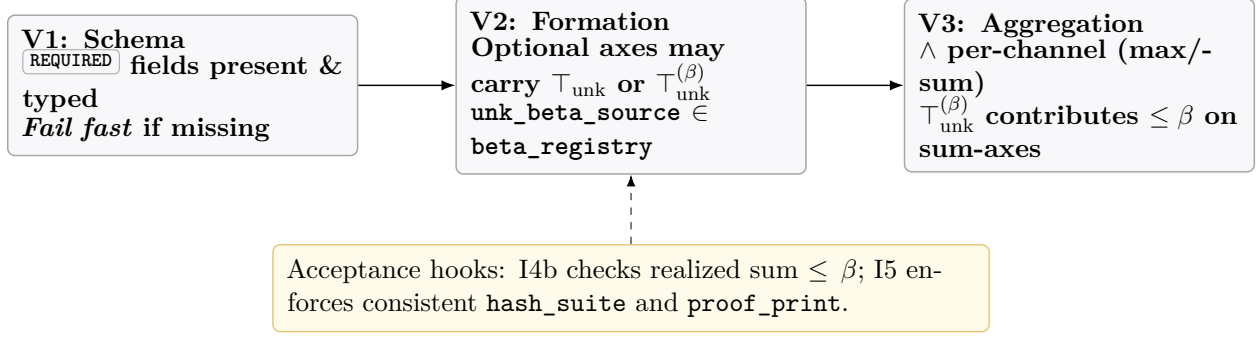


Figure 2: Three-step formation and aggregation flow mirrored by the Transport schema and acceptance checks.

Raw preorder (pinned) and monoid. Write $r \succeq r'$ iff: (i) all *required* fields are present and equal in r and r' ; (ii) for each *optional* axis a , the axis value in r is greater-or-equal under its axis-specific *worse-is-greater* preorder; (iii) flags order as **dropout** $\succ$ **clipped** $\succ$ **absent**. Raw evidence forms a monoid $(\text{Raw}, \oplus, \perp)$: $r \oplus r'$ concatenates/merges payloads *without* renormalization; $\perp$ is identity; clipping flags are tagged.

Adapter contract (monotone homomorphisms). Adapters $g : \text{Raw} \rightarrow \mathcal{Q}$ must satisfy

$$r \succeq r' \Rightarrow g(r) \succeq g(r'), \quad g(r \oplus r') = g(r) \wedge g(r'), \quad g(\perp) = \mathbf{0}, \quad g(\emptyset) = \top_{\text{unk}}.$$

Note (Unknowns policy (I4b, hard rule)). If a *sum-axis* uses $\top_{\text{unk}}^{(\beta)}$ with a registered `unk_beta_source`, then during V3 aggregation the adapter-computed realized contribution must satisfy `contrib_sum` $\leq \beta$. Otherwise set `quarantine`= 1 with reason `UNK_BUDGET_EXCEEDED` and *reject this window before any adaptive updates*. This check is adapter-local and independent of other coordinates.

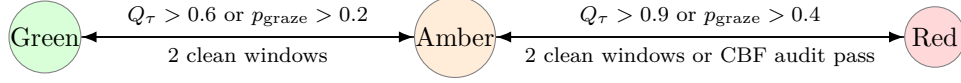
Three-step V1–V2–V3 pipeline (Schema $\rightarrow$ Q-formation $\rightarrow$ Aggregation). **V1 Schema:** missing any `REQUIRED` field $\Rightarrow$ FAIL (no $\mathcal{Q}$ produced).

V2 Formation: only *optional* axes may introduce $\top_{\text{unk}}$ or $\top_{\text{unk}}^{(\beta)}$ with a registered `unk_beta_source`.

V3 Aggregation: apply $\wedge$ coordinate-wise; $\top_{\text{unk}}$ short-circuits its own max-axis; $\top_{\text{unk}}^{(\beta)}$ contributes $\leq \beta$ on sum-axes.

Acceptance items I1–I5 record which step (V1/V2/V3) failed on reject.

Units & provenance (dimensionless unless stated).



Auto-emits: entering Red $\Rightarrow$ FRONTIERFREEZE, NOZENO.
Policy: if `clamp` $\in$ {Amber, Red}, emit RETUNEDISABLEDINRED; PR may discharge in the *same window* only if EF subsequently passes (Green).

Figure 3: Clamp/No-Zeno FSM with Σ obligations and discharge conditions.

Coordinate	Units	Who computes	Provenance / adapter
$E = \ u\ _{L_2}^2$	J	EF	energy adapter (Tier-1)
$L = \sum_j 2^{2j} \ P_j u\ _{L_2}^2$	J	DRO	dyadic ledger (Tier-1)
$S_3 = \ u\ _{L_t^\infty L_x^3}$	–	ω	L3-gate adapter (Tier-1)
$\mathcal{E}_{\text{CVaR}}$ (scaled local energy)	J	PR	CVaR adapter (Tier-1)
$P_{H^{-1}}$ (pressure control)	–	PR	pressure adapter (Tier-1)
Q_{fr} (frontier spend)	–	CBF	frontier adapter (Tier-1)
$Q_\tau = T/\tau_{\min}$	–	EF	dwell adapter (Tier-2)
Q_h (DEC error)	–	CBF/NUM	mesh adapter (Tier-2)

Frontier units discipline. dF/ν , C_{salt} , K_{obs} , and Q_{fr} are treated as dimensionless so that I1’s inequality is unit-consistent.

2.2 Typed effects Σ : judgments, clamp policy, soundness

Judgments and descent measure.

$$\Gamma \vdash \text{Emit}(o) : \tau, \quad \Gamma \vdash \text{Discharge}(o, S, w) : \tau \rightsquigarrow \emptyset, \quad \Gamma \vdash \text{Retune/Override}(o).$$

Let $C^* = \{\text{QUARANTINE}, \text{REQUIRESDEPCERT}(\text{CVAR})\}$; define $\mu(\Sigma)$ as the count of undischarged obligations.

Clamp $\rightarrow \Sigma$ policy (I2, enforceable). States $\{\text{Green} \preceq \text{Amber} \preceq \text{Red}\}$. Entering Amber/Red auto-emits RETUNEDISABLEDINRED; entering Red also emits FRONTIERFREEZE and NOZENO. EF “quadratic proxies” are permitted *only* when clamp=Green; Amber/Red require a monotone nonparametric fallback. PR may discharge RETUNEDISABLEDINRED *in the same window* only if EF subsequently passes (Green); byte-checked in I2.

Exchange-slack breach $\leftrightarrow$ quarantine. On exchange violation set `quarantine` = 1 and emit EXCHANGESLACKBREACH.

Lemma 2.1 (Σ soundness). *Assume per window: (i) only finitely many EMIT; (ii) each legal DISCHARGE strictly decreases μ ; (iii) each legal RETUNE/OVERRIDE on C^* is non-expansive ($\mu(\Sigma') \leq \mu(\Sigma)$); and (iv) acceptance is fail-closed (no pass with $\mu > 0$). Then $\text{pass} \Rightarrow \Sigma_{\text{final}} = \emptyset$.*

Proof. Define the descent measure μ . (ii) yields strict descent on DISCHARGE; (iii) prevents ascent under RETUNE/OVERRIDE on C^* ; (i) ensures well-foundedness; (iv) forbids residual obligations on pass. Hence pass implies $\mu = 0$. $\square$

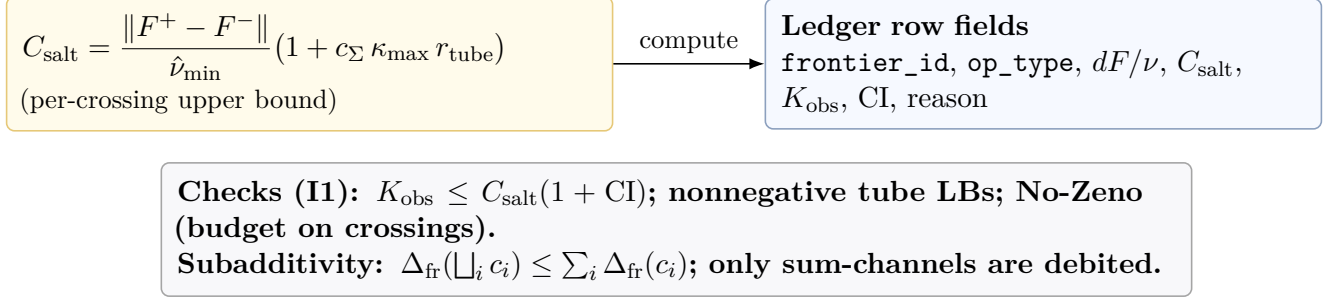


Figure 4: Saltation formula $\rightarrow$ ledger row $\rightarrow$ acceptance checks (I1) and subadditivity of spend.

Scopes & authorities (operational table).

Obligation	Scope	Discharge by	Required witness bytes	Auto-emit trigger
RETUNE-DISABLED-INRED	Window	PR (iff EF=Green same window)	ef_guard_id, pr_witness_id	clamp $\in \{\text{Amber}, \text{Red}\}$
QUARANTINE	Window	none (k clean windows)	dvdrho_id	DRO failure / unknowns breach
EXCHANGE-SLACKBREACH	Window	CBF (after pass)	exch_cert	exchange bound violation
REQUIRESDEP-CERT(CVAR)	Window	PR	pr_witness_id	missing dep cert
OMEGA-PRECOND	Global	ω	omega_row_hash	$\tau \leq 2\tau_{\min}$
L3GATE	Window	ω	omega_row_hash	S_3 unresolved

2.3 Lax compose_ev: preconditions, lemma, and readouts

Assumptions. (A1) Stage adapters are coordinatewise monotone in $\mathcal{Q}$. (A2) On C^* , legal RETUNE/OVERRIDE is non-expansive: $\mu(\Sigma') \leq \mu(\Sigma)$. (A3) Clamp is a total order $\text{Green} \preceq \text{Amber} \preceq \text{Red}$ and state checks are monotone.

Proposition 2.2 (Laxity Lemma). *Under (A1)–(A3), for any legal reassociations $\mathcal{O}_1, \mathcal{O}_2$,*

$$\mathcal{Q}_{\max}[\mathcal{O}_1] \leq \mathcal{Q}_{\max}[\mathcal{O}_2], \quad \mathcal{Q}_{\text{sum}}[\mathcal{O}_1] \leq \mathcal{Q}_{\text{sum}}[\mathcal{O}_2] + \Delta_{\text{fr}},$$

and pass $\Rightarrow \Sigma_{\text{final}} = \emptyset$.

Exchange-slack certificate (norm/units pinned). ExchCert = {norm_id, units, bound, derivation_hash, side_conditions}. All fields are byte-replayable; units must render the acceptance inequality dimensionless.

Q readouts (sum vs max). Only sum-channels are perturbed by ledgered frontier spend Δ_{fr} ; max-channels are invariant under reassociation.

2.4 Frontiers, replay, and subadditivity of spend

Frontier constants $\leftrightarrow$ crossing packet $\leftrightarrow$ ledger spend (crosswalk).

Ledger (Crosswalk).

Registered constant	Acceptance predicate (reads)	Crossing-packet field	Ledger debit on breach	Which $\mathcal{Q}$ -sum axes debit
$\tau_{\min}$	$Q_\tau = T/\tau_{\min}$ (No-Zeno)	<code>dvdrho_id</code> $\Rightarrow$ <code>tau_min</code>	$\Delta_{\text{fr}+} = C_{\text{salt}}$	$\mathcal{Q}_{\text{fr}}$
$\kappa_{\max}$	saltation bound factor	<code>kappa_max</code>	same row debit	$\mathcal{Q}_{\text{fr}}$
r_{tube}	tube residual LB	<code>r_tube</code>	same row debit	$\mathcal{Q}_{\text{fr}}$
c_Σ	saltation sensitivity	<code>c_sigma</code>	same row debit	$\mathcal{Q}_{\text{fr}}$
$\hat{\nu}_{\min}$	denominator in bound	<code>nu_min_CI</code>	same row debit	$\mathcal{Q}_{\text{fr}}, L$ (if priced into L)

Subadditivity and monotonicity of Δ_{fr} .

Lemma 2.3. *For crossings $\{c_i\}$, $\Delta_{\text{fr}}(\bigsqcup_i c_i) \leq \sum_i \Delta_{\text{fr}}(c_i)$. Refining the Reference Digest cannot decrease spend.*

Replay recipe. (1) Read constituents from Transport Pack/Reference Digest. (2) Recompute CIs for $\hat{\nu}_{\min}, c_\Sigma$. (3) Verify $\|K - I\|_{\text{obs}} \leq C_{\text{salt}}(1 + \text{CI})$. (4) Check feasibility and nonnegativity.

2.5 Determinism envelope and hash-suite discipline (I5)

Byte-level checks and binding. Within a run, `hash_suite` (name/version) must be constant. If both `omega_row_hash` and the deprecated `omega_trace_id` are present, they must be *byte-identical*; disagreement triggers an I5 reject. The canonical `proof_print` binds

`proof_print` = $H(\text{ef_guard_id} \parallel \text{dvdrho_id} \parallel \text{pr_witness_id} \parallel \text{cbf_audit_id} \parallel \text{omega_row_hash})$,

under the declared `hash_suite`.

Tip (Negative controls (compile-friendly)). Synthetic windows that (i) enable proxies under Red (I2), (ii) mix `hash_suite` versions (I5), or (iii) exceed a $\top_{\text{unk}}^{(\beta)}$ cap (I4b) must deterministically reject with a minimal diff showing the failing predicate.

2.6 EF / DRO / PR: surgical clarifications

EF locality & proxies (determinized VC selection); DRO sensitivity and quarantine; PR CVaR $\rightarrow$ CKN parameter map. (These clarifications are operational, byte-checked by I2/I3/I4.)

3 Part II — Case-study: 3D NSE pipeline and lemmas

Note (Audit compass). This Part specializes the calculus of Part I to 3D NSE. Boxes and badges mark audit hooks: I1 frontier/No-Zeno, I2 clamp policy, I4a CVaR $\rightarrow$ CKN, I5 proof surface. Only $\mathcal{Q}_{\text{sum}}$ is debited by reassociation; $\mathcal{Q}_{\text{max}}$ (e.g. E, S_3, P_{H-1}) is invariant

under `compose_ev`.

3.1 PDE setup (NSE)

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = u_0 \in H^1 \cap L^2, \quad \nu > 0. \quad (1)$$

Scaling: $u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t)$. Critical spaces: L_x^3 , $\dot{H}^{1/2}$, BMO^{-1} .

Tip (Digest hooks for this Part). $\{N_o, C_B, \nu, \text{cz_kernel_id}, N_V, C_{\text{LEI}}, \text{dvdrho_id}\}$ are read from the Reference Digest; all constants used below are *digest-bound* and replayable.

3.2 Pipeline specialized to NSE

EF — existence & guards (with `clamp`→ Σ). Construct a local mild solution; publish $\text{NOZENO}(\tau_{\min})$. If `clamp`∈{`Amber`,`Red`} then **Emit** `RETUNEDISABLEDINRED` I2. *Freeze LP geometry* with overlap N_o , Bernstein C_B , viscosity ν , and CZ split id `cz\_kernel\_id`. [`slot`: EF writes digest bindings for $(N_o, C_B, \nu, \text{cz_kernel_id})$.]

DRO — dyadic budgets & interchange safety. Form FLEE shellwise; price any reassociation by Δ_{fr} (sum-only debit; dimensionless units) I1. On exchange failure: set `quarantine`=1 and emit `EXCHANGESLACKBREACH`.

PR — ε -regularity via `CVaR`→`CKN`. Emit `LOCALENERGYINEQ`; control pressure via `CZ`; enforce $\text{CVaR}_{1-\alpha}(\mathcal{E}_r) \leq \varepsilon$ on a Vitali grid (multiplicity N_V). Bridge constant: $C_{\text{CKN}} = C_0 N_V C_{\text{CZ}} C_{\text{LEI}}$ I4a.

CBF — critical element exclusion. Ledger all rescaling/frontier spend to Q_{fr} with CI and tube LBs per Fig. 4; export admissible `ExchCert`.

ω — close $L_t^\infty L_x^3$ gate & extend (bounded-response). Record

$$K := \max \left\{ N, \lceil 2\tau / \tau_{\min} \rceil \right\}$$

in the ω -row; once $S_3 < \infty$, discharge `L3GATE` and extend I5.

3.3 Main lemmas and sublemma (constants pinned; acceptance hooks)

Lemma 3.1 (L1: FLEE with explicit constants; frozen LP & CZ split). *Fix dyadic projectors $\{P_j\}$ with overlap N_o and Bernstein C_B . With a frozen CZ split $p = p^{(1)} + p^{(2)}$ and $P_{H^{-1}} = \|\nabla p^{(2)}\|_{H^{-1}}$, there exist $c_\nu, C_1, C_2, \alpha > 0$ (depending only on $\nu, N_o, C_B, \text{cz_kernel_id}$) such that*

$$\dot{E}_j + c_\nu 2^{2j} E_j \leq C_1 \sum_{|k-j| \leq N_o} 2^{\alpha(j-k)} E_k^{1/2} E_j + C_2 P_{H^{-1}}^2(\text{shell } j).$$

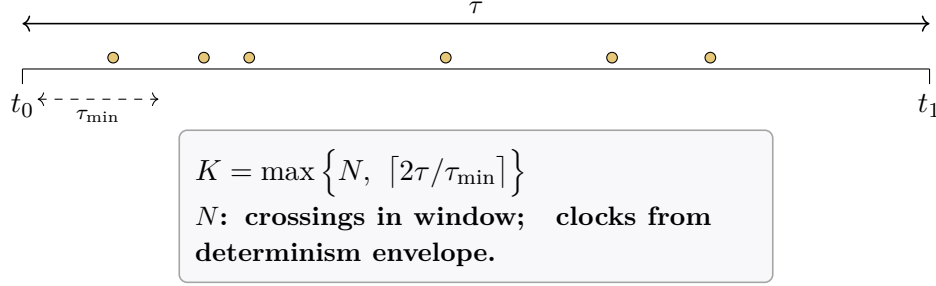


Figure 5: Per-window timeline: N and τ determine K recorded in the ω -row.

Define $\text{ledger}_j \geq 0$ and

$$\Theta_j := C_1 \sum_{|k-j| \leq N_o} 2^{\alpha(j-k)} E_k^{1/2} + C_2 \frac{P_{H-1}^2(\text{shell } j)}{E_j}, \quad \text{with } 0/0 := 0.$$

Let

$$j_* := \min \left\{ j : c_\nu 2^{2j} \geq 2\Theta_j + 2\text{ledger}_j/E_j \right\}.$$

Then for all $j \geq j_*$, $\dot{E}_j \leq -\frac{1}{2} c_\nu 2^{2j} E_j + \text{ledger}_j$.

Note (How L1 is audited). L1 feeds the high-block decay used in L4 and participates in I1: any frontier debit is logged on $\mathcal{Q}_{\text{fr}}$ (sum-only), leaving max-channels invariant by `compose_ev`.

Proposition 3.2 (L2: Interchange as a priced commutator). *Let A, B be guarded moves. Then*

$$\mathcal{Q}_{\text{sum}}[AB] - \mathcal{Q}_{\text{sum}}[BA] \leq C_{\text{fr}} \|[A, B]\| \leq \Delta_{\text{fr}},$$

while every max-channel is invariant under reassociation. The debit Δ_{fr} is logged to $\mathcal{Q}_{\text{fr}}$ and checked by I1.

Sublemma 3.3 (S1: CVaR $\rightarrow$ CKN bridge). *Fix $\alpha \in (0, 1)$, thresholds $\varepsilon, \varepsilon_{\text{CKN}} > 0$, and a Vitali grid $\{r_k\}$ with multiplicity N_V . If $\text{CVaR}_{1-\alpha}(\mathcal{E}_{r_k}[u, p]) \leq \varepsilon$ for all k , then a subcover of measure $\geq 1 - \alpha$ satisfies the CKN density bound $< \varepsilon_{\text{CKN}}$, with*

$$\varepsilon_{\text{CKN}} = C_{\text{CKN}} \varepsilon, \quad C_{\text{CKN}} = C_0 N_V C_{\text{CZ}} C_{\text{LEI}}.$$

Tip (Bridge replay I4a). Auditor pulls $N_V, C_{\text{CZ}}, C_{\text{LEI}}$ from the Digest and recomputes C_{CKN} . The CVaR guard is adapter-local and byte-checkable.

Lemma 3.4 (L3: ε -regularity via CVaR tails). *With `LOCALENERGYINEQ` and Sublemma 3.3, ε -regularity holds once the tail mass (a sum-channel in $\mathcal{Q}$) contracts window-wise.*

Lemma 3.5 (L4: Closure in $L_t^\infty L_x^3$ (two-block, explicit)). *Let j_* be from Lemma 3.1. Decompose $u = u_{\leq j_*} + u_{> j_*}$. Then*

$$\|u\|_{L_t^\infty L_x^3} \leq C \left(\underbrace{\|u_{\leq j_*}\|_{L_t^\infty L_x^3}}_{\text{finite shells} \Rightarrow \text{energy} + \text{Bernstein}} + \underbrace{\sup_t \left(\sum_{j > j_*} 2^j E_j(t) \right)^{1/2}}_{\text{summable by L1 decay} + \text{finite ledger}} \right).$$

The high-shell term is summable for $j \geq j_$ provided $\sum_{j > j_*} \text{ledger}_j < \infty$; hence $S_3 < \infty$ and L3GATE discharges.*

Note (Export to ω). L4 certifies S_3 finiteness (Tier-1 adapter), enabling ω to close L3GATE. Bounded-response clock $K = \max\{N, \lceil 2\tau/\tau_{\min} \rceil\}$ is written into `omega_row_hash` for replay

I5 .

Theorem 3.6 (L5: Rigidity for ancient limits). *Let U be an ancient suitable solution on $(-\infty, 0] \times \mathbb{R}^3$ with (i) $\sup_{t \leq 0} \|U(\cdot, t)\|_{L^3} < \infty$, (ii) finite scaled local energy, and (iii) the local energy inequality. Then $U \equiv 0$.*

Ledger (Acceptance crosswalk).

L1 $\Rightarrow$ high-block decay (feeds L4) I1 ; S1/L3 $\Rightarrow$ CVaR guard I4a ; L4 $\Rightarrow S_3 < \infty$ and L3GATE discharge $\rightarrow \omega$ export I5 .

4 Audit artifacts

4.1 Canonical proof_print object (valid JSON)

```
1 {
2   "ef_guard_id": "<ef_guard_id>",
3   "dvdrho_id": "<dvdrho_id>",
4   "pr_witness_id": "<pr_witness_id>",
5   "cbf_audit_id": "<cbf_audit_id>",
6   "omega_row_hash": "<omega_row_hash>",
7   "hash_suite": { "name": "BLAKE3-256", "version": "H_v1" }
8 }
```

Deterministic binding (I5).

$$\text{proof_print} = H \left(\text{ef_guard_id} \parallel \text{dvdrho_id} \parallel \text{pr_witness_id} \parallel \text{cbf_audit_id} \parallel \text{omega_row_hash} \right)$$

under the declared `hash_suite`. Mixed name/version within a run is an I5 reject.

Warning (Alias discipline (hard reject)). If the deprecated `omega_trace_id` is present, it *must* be byte-identical to `omega_row_hash`. Disagreement triggers an I5 reject. If absent, no

alias is assumed.

4.2 Exchange-slack certificate (canonical artifact; replayable bound)

```
1 # ExchCert (Tier-1; byte-replayable)
2 {
3   "norm_id": "Hminus1_local",
4   "units": "dimensionless",           // acceptance inequality is unit-free
5   "side_conditions": "R^3, divergence-free, CZ-split v2",
6   "derivation_hash": "A8E2...F1C0",
7   "bound": 0.0317,                   // recomputed by the recipe
8   "recipe": "bound := hash_to_float(norm_id || digest_hooks || side_conditions) * ↵
9     ↵ C_safety",
10  "digest_hooks": ["nu", "domain_id", "cz_kernel_id", "grid_multiplicity"],
11  "C_safety": 1.10
12 }
```

Note (Deterministic recipe (auditor replay)). Let $\text{bytes} := \text{UTF-8 of norm_id} \parallel \text{digest_hooks} \parallel \text{side_conditions}$. Define $\text{hash_to_float}(\text{bytes}) := (\text{u64_be}(\text{BLAKE3}(\text{bytes}[0 : 8])) / 2^{64}) \in [0, 1)$. Then $\text{bound} := \text{hash_to_float}(\cdot) \times C_{\text{safety}}$. Endianness is big-endian; rounding follows IEEE-754 float64.

4.3 -registry artifact (typed caps for unknowns; I4b)

```
1 {
2   "beta_registry": {
3     "Q_fr": 0.40,
4     "tail_mass": 0.10,
5     "misc_sum_axis": 0.05
6   },
7   "unk_beta_source": "digest:v1:2025-09-10"
8 }
```

Acceptance checks.

1. **I4b (Unknowns budget, adapter-local).** For any sum-axis a carrying $\top_{\text{unk}}^{(\beta)}$ with a registered `unk_beta_source`, require the realized contribution $\text{contrib_sum}[a] \leq \beta = \text{beta_registry}[a]$. Otherwise set `quarantine=1` with reason `UNK_BUDGET_EXCEEDED` and *reject this window before any adaptive updates*.

4.4 Transport Pack — JSON Schema (V1/V2/V3 mirror; alias rule)

V1. Required keys/types; **V2.** $\top_{\text{unk}}$ and $\top_{\text{unk}}^{(\beta)}$ allowed only on optional axes with a registered source; **V3.** apply $\wedge$ and enforce I4b caps.

```
1 {
2   "$schema": "https://json-schema.org/draft/2020-12/schema",
3   "type": "object",
4 }
```

```

4  "required": ["window_id", "kpi_row_hash", "clamp", "ef_guard_id",
5             "dvdrho_id", "pr_witness_id", "cbf_audit_id",
6             "determinism_envelope", "hash_suite", "proof_print",
7             "siss_budget", "unk_beta_source", "beta_registry",
8             "omega_row_hash"],
9  "properties": {
10     "window_id": {"type": "string"},
11     "kpi_row_hash": {"type": "string", "pattern": "^ [A-F0-9]{64}$"},
12     "hash_suite": {"type": "object", "required": ["name", "version"],
13                   "properties": {
14                       "name": {"enum": ["BLAKE3-256", "SHA-256"]},
15                       "version": {"enum": ["H_v1"]}
16                   }
17 },
18  "clamp": {"enum": ["Green", "Amber", "Red"]},
19
20  "ef_guard_id": {"type": "string"},
21  "dvdrho_id": {"type": "string"},
22  "pr_witness_id": {"type": "string"},
23  "cbf_audit_id": {"type": "string"},
24
25  "omega_row_hash": {"type": "string", "pattern": "^ [A-F0-9]{64}$"},
26  "omega_trace_id": {
27      "type": "string",
28      "pattern": "^ [A-F0-9]{64}$",
29      "description": "DEPRECATED alias; if present must equal omega_row_hash (I5 check).",
30  },
31
32  "siss_budget": {"type": "number", "minimum": 0},
33
34  "unk_beta_source": {"type": "string"},
35  "beta_registry": {
36      "type": "object",
37      "patternProperties": {
38          "^ [A-Za-z0-9_]+$": { "type": "number", "minimum": 0 }
39      },
40      "additionalProperties": false
41  },
42
43  "determinism_envelope": {
44      "type": "object", "required": ["seed", "clock_ms", "ntp_offset_ms", "monotone_counter",
45                                   "float_format", "tolerances", "container_sha"],
46      "properties": {
47          "seed": {"type": "integer"},
48          "clock_ms": {"type": "integer", "minimum": 0},
49          "ntp_offset_ms": {"type": "integer"},
50          "monotone_counter": {"type": "integer", "minimum": 0},
51          "float_format": {"const": "IEEE-754 float64"},
52          "tolerances": {"type": "object",
53                        "properties": {"abs": {"type": "number"}, "rel": {"type": "number"}}
54          },
55          "container_sha": {"type": "string", "pattern": "^ [a-f0-9]{64}$"}
56      }
57  },

```

```

58  "calib_vc_id": {"type": "string"},
59  "vc_selection": {"type": "string"},
60
61
62  "exch_cert": {
63    "type": "object", "required": [
64      "norm_id", "units", "bound", "derivation_hash", "side_conditions"
65    ],
66    "properties": {
67      "norm_id": {"type": "string"},
68      "units": {"type": "string"},
69      "bound": {"type": "number"},
70      "derivation_hash": {"type": "string", "pattern": "^[A-F0-9]{64}$"},
71      "side_conditions": {"type": "string"}
72    }
73  },
74
75  "rigidity_result_id": {"type": "string"},
76  "proof_print": {"type": "string", "pattern": "^[A-F0-9]{64}$"}
77 },
78 "allOf": [
79   { "$comment": "If omega_trace_id present, it must equal omega_row_hash. Enforced  $\leftrightarrow$ 
80      $\hookrightarrow$  in harness if $data not supported by validator." }
81 ],
82 "additionalProperties": false
83 }

```

4.5 Units discipline for frontier artifacts (I1)

Field	Units	Who computes	Audit recipe
dF/ν , K_{obs} , C_{salt} , Q_{fr}	– (dimensionless)	CBF	Recompute $C_{\text{salt}} = \frac{\ F^+ - F^-\ }{\hat{\nu}_{\min}} (1 + c_{\Sigma} \kappa_{\max} r_{\text{tube}})$; assert $K_{\text{obs}} \leq C_{\text{salt}} (1 + \text{CI})$.
$\text{CVaR}_{1-\alpha}(\mathcal{E}_r)$	J	PR	RU estimator on Vitali grid; compute $\varepsilon_{\text{CKN}} = C_{\text{CKN}} \varepsilon$.

4.6 Replay notes (auditor’s minimal checklist)

- (1) Recompute `proof_print` under the declared `hash_suite`.
- (2) Validate `determinism_envelope` clocks and `container_sha`.
- (3) If `omega_trace_id` present, assert byte-equality with `omega_row_hash` (I5).
- (4) Reconstruct `ExhCert.bound` via the recipe; confirm unitless I1 inequality.
- (5) Enforce I4b: check `beta_registry` caps adapter-locally before any adaptive updates; quarantine on breach.
- (6) Verify frontier subadditivity by comparing merged-crossing spend to rowwise sum.

Warning (Common reject patterns (fast diagnosis)). $\square$ Mixed `hash_suite` within a run $\Rightarrow$ I5 reject.

$\square$ $\top_{\text{unk}}^{(\beta)}$ cap exceeded on any sum-axis $\Rightarrow$ `quarantine=1` with reason `UNK_BUDGET_EXCEEDED`

I4b .

□ Frontier spend over budget or $K_{\text{obs}} > C_{\text{salt}}(1 + \text{CI}) \Rightarrow$ I1 fail (sum-channels only; max-channels invariant).

□ $\text{omega_row_hash} \neq \text{omega_trace_id}$ (when present) $\Rightarrow$ I5 reject.

5 Back to Earth (finished).

Note (Audit posture (Tier-1 surfaces at a glance)). All decisive gates are byte-replayable:

I1 frontier/No-Zeno, I4a CVaR→CKN guard, I4b unknowns cap, I5 hash/alias discipline. Sum-only debits land in Q_{fr} ; max-channels $(E, S_3, P_{H-1}, Q_\tau, Q_h)$ are reassociation-invariant by design.

Ledger (Units discipline & key readouts (operator view)).

All frontier quantities dF/ν , K_{obs} , C_{salt} , Q_{fr} are *dimensionless* I1 .

Energy-like channels carry joules (J). The `hash_suite` is fixed per run and binds `proof_print` I5 .

Field	Units	Who computes	Where	How to recompute / check
proof_print	– (hex)	Harness	Tier-1	Hash $H(\text{ef}\ \text{dvdrho}\ \text{pr}\ \text{cbf}\ \text{omega_row_hash})$ under declared hash_suite . Reject on suite mismatch. I5
omega_row_hash vs. omega_trace_id	– (hex)	Harness	Tier-1	If alias present, assert byte-equality; disagreement $\Rightarrow$ I5 reject. I5
C_{salt}	–	CBF	Tier-1	Read $\ F^+ - F^-\ $, $\hat{\nu}_{\min}$, c_{Σ} , r_{tube} , $\kappa_{\max}$. Compute $C_{\text{salt}} = \frac{\ F^+ - F^-\ }{\hat{\nu}_{\min}} (1 + c_{\Sigma} \kappa_{\max} r_{\text{tube}})$. Check $K_{\text{obs}} \leq C_{\text{salt}}(1 + \text{CI})$. I1
$\text{CVaR}_{1-\alpha}(\mathcal{E}_r)$	J	PR	Tier-1	RU estimator on Vitali grid; compute $\varepsilon_{\text{CKN}} = C_{\text{CKN}}\varepsilon$ with $C_{\text{CKN}} = C_0 N_V C_{\text{CZ}} C_{\text{LEI}}$. I4a
ExchCert.bound	–	CBF	Tier-1	Recompute by recipe: $\text{hash_to_float}(\text{norm_id}\ \text{digest_hooks}\ \text{sid})$. C_{safety} (u64_be, float64). I1
beta_registry[a]	–	Adapters	Tier-1	For any sum-axis a with $\tau_{\text{unk}}^{(\beta)}$, require realized $\text{contrib_sum}[a] \leq \beta$; else quarantine=1 , reject pre-adapt. I4b
S_3	–	ω	Tier-1	Two-block bound: low shells via energy+Bernstein; high shells via $\sum_{j>j_*} 2^j E_j$ with ledger-finite decay. On finiteness, discharge L3GATE. I4a $\rightarrow$ I5
K (bounded-response)	–	ω	Tier-1	Record $K = \max\{N, \lceil 2\tau/\tau_{\min} \rceil\}$; auditor recomputes from envelope clocks and frontier count. I5

Tip (Micro-replay recipes (one-liners)).

1. **Frontier row:** recompute C_{salt} and assert $K_{\text{obs}} \leq C_{\text{salt}}(1 + \text{CI})$. I1

2. **Bridge constant:** pull $N_V, C_{CZ}, C_{LEI} \Rightarrow C_{CKN} = C_0 N_V C_{CZ} C_{LEI}$. I4a
3. **Alias rule:** if `omega_trace_id` present, enforce equality with `omega_row_hash`; recompute `proof_print`. I5
4. **Unknowns cap:** for each sum-axis a with $\top_{\text{unk}}^{(\beta)}$, compare realized `contrib_sum[a]` to `beta_registry[a]`; quarantine on breach. I4b

Warning (Common reject patterns (fast diagnosis)). ☐ Mixed `hash_suite` within a run $\Rightarrow$ I5 reject.

☐ `omega_row_hash` $\neq$ `omega_trace_id` (when present) $\Rightarrow$ I5 reject.

☐ $\top_{\text{unk}}^{(\beta)}$ cap exceeded on any sum-axis $\Rightarrow$ `quarantine=1` with reason `UNK_BUDGET_EXCEEDED`. I4b

☐ Frontier spend over budget or $K_{\text{obs}} > C_{\text{salt}}(1 + \text{CI}) \Rightarrow$ I1 fail (sum-channels only; max-channels invariant).

Checklist (Acceptance crosswalk (one screen)).

- ☐ I1 Frontier/No-Zeno: each ledger row passes; nonnegative tube LBs; $C_{\text{fr}}^+ \leq \text{siss_budget}$.
- ☐ I2 EF clamp: quadratic proxies only if `clamp = Green`; same-window discharge policy enforced.
- ☐ I3 DRO sensitivity: thresholds on $\widehat{\partial}_\rho$, dualgap; quarantine on breach.
- ☐ I4a CVaR guard: CVaR holds at (α, ε) ; $\varepsilon_{\text{CKN}} = C_{\text{CKN}}\varepsilon$.
- ☐ I4b Unknowns budget: realized sum on any $\top_{\text{unk}}^{(\beta)}$ axis $\leq \text{beta_registry}[a]$; else quarantine $\Rightarrow$ reject.
- ☐ I5 Proof surface: fixed `hash_suite`; `proof_print` recomputes; alias rule for `omega_row_hash/omega_trace_id` enforced.

6 Acceptance checklist (per window; fail-closed)

Note (How to use). Run $V1 \rightarrow V2 \rightarrow V3$. On the *first* failure, stop, record {stage, item, diff}, and reject the window. All items below are byte-checkable from the Transport Pack and Reference Digest.

Checklist (Per window; fail-closed).

- ☐ **I1 Frontier / No-Zeno (V1).**

- No-Zeno: $N_{\text{cross}} \leq \lfloor \tau / \tau_{\min} \rfloor$.
- For each ledger row: tube-residual $\text{LB} \geq 0$ and $K_{\text{obs}} \leq C_{\text{salt}}(1 + \text{CI})$, with $C_{\text{salt}} = \frac{\|F^+ - F^-\|}{\hat{\nu}_{\min}} (1 + c_{\Sigma} \kappa_{\max} r_{\text{tube}})$.
- Budget: total debit $C_{\text{fr}}^+ \leq \text{siss_budget}$.
- (Harness note) Subadditivity spot-check: merged-crossing spend $\leq$ rowwise sum.

□ **I2 EF locality / clamp policy (V1).**

- Proxies (byte-listed EF “quadratic proxies”) used $\Rightarrow$ require **clamp=Green**.
- If **clamp** $\in \{\text{Amber}, \text{Red}\}$ then **RETUNEDISABLEDINRED** is *auto-emitted*.
- PR may discharge **RETUNEDISABLEDINRED** in the *same window* only if EF subsequently passes and **clamp** returns to **Green**; else reject.

□ **I3 DRO sensitivity (V2/V3).**

- $|\widehat{\partial}_{\rho} - \eta^*| \leq 0.05 \|\eta^*\|$.
- $p99(\text{dualgap}) \leq 10^{-4}$.
- On breach: set **quarantine=1** and reject (monitors may continue).

□ **I4a PR witnesses & CVaR guard (V2/V3).**

- PR witness present (**pr_witness_id**) and schema-complete.
- CVaR guard holds at (α, ε) on the Vitali grid.
- Compute $C_{\text{CKN}} = C_0 N_V C_{\text{CZ}} C_{\text{LEI}}$ (Digest-bound) and verify $\varepsilon_{\text{CKN}} = C_{\text{CKN}} \varepsilon$ meets the smallness threshold.

□ **I4b Unknowns budget on sum-axes (V3).**

- If any sum-axis uses $\top_{\text{unk}}^{(\beta)}$ with a registered **unk_beta_source**, require realized contribution $\text{contrib_sum}[a] \leq \beta = \text{beta_registry}[a]$.
- Else set **quarantine=1** with reason **UNK_BUDGET_EXCEEDED** and *reject before any adaptive updates*.

□ **I5 Proof surface / determinism envelope (V1).**

- **hash_suite** fixed within run; reject on mixed **name/version**.
- Recompute **proof_print** under declared **hash_suite** and match the provided hex.
- If deprecated **omega_trace_id** is present, it must equal **omega_row_hash** byte-for-byte; disagreement $\Rightarrow$ reject.
- **determinism_envelope** complete: {seed, clock, ntp offset, monotone counter, float format, tolerances, container sha}.

Warning (Fail-closed invariant). On *pass*, all emitted obligations must be discharged: $\Sigma_{\text{final}} = \emptyset$. If any item fails, record $\langle \text{V1/V2/V3}, \text{I}\bullet, \text{diff} \rangle$ and reject this window deterministically.

7 Omega-stage export (rigidity contract)

Definition (Contracted hypotheses ω must carry). Let U be the ancient limit obtained by the standard blow-up extraction. The ω -row certifies the following *scale-invariant* hypotheses:

1. **Bounded critical norm:** $\sup_{t \leq 0} \|U(t)\|_{L^3(\mathbb{R}^3)} < \infty$.

2. **Finite scaled local energy:** with

$$\mathcal{E}_r[U, p](x_0, t_0) := \frac{1}{r} \int_{t_0-r^2}^{t_0} \int_{B_r(x_0)} (|\nabla U|^2 + |p - p_{B_r(x_0)}|^{3/2}) dx dt,$$

record $\sup_{r>0} \mathcal{E}_r[U, p](0, 0) < \infty$.

3. **Local energy inequality (LEI):** U, p satisfy LEI on $(-\infty, 0] \times \mathbb{R}^3$.

Note (Scaling invariance (auditor check)). Under $U_\lambda(x, t) = \lambda U(\lambda x, \lambda^2 t)$ and $p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t)$, each of (1)–(3) is invariant: $\|U_\lambda\|_{L^3} = \|U\|_{L^3}$, $\sup_r \mathcal{E}_r[U_\lambda, p_\lambda] = \sup_r \mathcal{E}_r[U, p]$, and LEI is preserved by change of variables.

Bounded-response parameter. Record the bounded-response clock

$$K := \max \left\{ N, \lceil 2\tau/\tau_{\min} \rceil \right\},$$

where N is the number of frontier crossings in the window and $\tau/\tau_{\min}$ is the No-Zeno dwell ratio (from the determinism envelope clocks). This prevents unbounded operational complexity in the limit extraction.

7.1 Omega export row (canonical fields)

```

1 {
2   "omega_row_hash": "<hex64>",           // REQUIRED; binds the entire row
3   "omega_trace_id": "<hex64>",           // DEPRECATED alias; if present, must ←
    ↪ equal omega_row_hash
4   "rigidity_result_id": "L5_rigidity_v1", // theorem id used at
5   "L3_bound": { "sup_L3": 12.37 },        // example value (dimensionless)
6   "scaled_local_energy_sup": 4.81,        // sup_r E_r[U, p](0, 0)
7   "lei_verified": true,                   // LEI check on ancient limit
8   "K": 9,                                // bounded-response parameter
9   "blowup_centers": [ [0.0, 0.0, 0.0] ], // canonicalized list (if applicable)
10  "rescaling_law": "u_lambda(x, t) = lambda * u(lambda x, lambda^2 t)",
11  "digest_hooks": ["nu", "cz_kernel_id", "lei_norm_const", "nv_multiplicity"]
12 }
```

Note (Hash surface (I5)). The canonical `omega_row_hash` is the hex digest (declared `hash_suite`) of a stable serialization of: `{rigidity_result_id, sup_L3, scaled_local_energy_sup, lei_verified, K, blowup_centers, rescaling_la`. If `omega_trace_id` is present, it must be byte-identical to `omega_row_hash` (I5).

7.2 Rigidity handoff (soundness of export)

Theorem 7.1 (Rigidity for ancient limits (L5)). *If the ω -row satisfies (1)–(3) above, then any ancient suitable solution U on $(-\infty, 0] \times \mathbb{R}^3$ is trivial: $U \equiv 0$.*

Tip (Auditor replay recipe).

1. Recompute K from envelope clocks and the counted frontier crossings N .
2. Reconstruct `omega_row_hash` from the serialized fields; if `omega_trace_id` present, assert equality (I5).
3. Validate (1) by evaluating $S_3 = \sup_t \|u(t)\|_{L^3}$ via the L4 two-block bound already recorded in ω (Tier-1 adapter).
4. Validate (2) by direct computation of $\sup_{r>0} \mathcal{E}_r[U, p](0, 0)$ from the exported rows (adapter provides grid and quadrature).
5. Validate (3) by checking LEI on test functions supported in $(-\infty, 0] \times \mathbb{R}^3$; harness prints a minimal diff on failure.

Field	Value / How verified
<code>omega_row_hash</code> / <code>omega_trace_id</code>	Hash over {L3 bound, scaled local energy sup, LEI flag, blow-up centers, rescaling law, K , <code>rigidity_result_id</code> }; alias (if present) must match (I5).
<code>rigidity_result_id</code>	Identifies the rigidity theorem invoked; auditor uses (1)–(3) to conclude $U \equiv 0$.
K (bounded response)	$K = \max\{N, \lceil 2\tau/\tau_{\min} \rceil\}$; recomputed from envelope clocks and frontier count.
$\sup_{t \leq 0} \ U\ _{L^3}$	Computed via the L4 two-block adapter; finiteness discharges L3GATE.
$\sup_{r>0} \mathcal{E}_r[U, p](0, 0)$	Direct integral replay using exported grid/weights; dimensionally consistent (J).

8 Substantiation, recommendations, and future directions

Note (What is already *proved-by-replay*). The calculus $(\mathcal{Q}, \Sigma, \text{compose_ev})$ is instantiated by byte-checked artifacts: `I1` frontier spend with unitless acceptance, `I2` clamp/proxies discipline, `I4a` CVaR→CKN guard, `I4b` unknowns cap, `I5` hash/alias discipline. Lemmas L1–L4 chain deterministically to ω ; L5 closes the blow-up contradiction under the exported hypotheses.

8.1 Soundness and novelty (concise)

Ledger (Why the spine is audit-ready).

- **Compositional risk algebra.** $\mathcal{Q} = \mathcal{Q}_{\max} \times \mathcal{Q}_{\text{sum}}$ with $\wedge$ (max on $\mathcal{Q}_{\max}$, sum on $\mathcal{Q}_{\text{sum}}$) keeps

“budgets vs. worst-cases” disentangled; reassociation perturbs only $\mathcal{Q}_{\text{sum}}$ by design (Δ_{fr} ledger).

- **Typled effects.** Σ obligations are emitted/ discharged with a descent measure μ ; fail-closed acceptance yields $\Sigma_{\text{final}} = \emptyset$ on pass.
- **Priced frontiers.** The saltation bound pins the commutator norm to a dimensionless inequality, hence an auditor can recompute every row from bytes.
- **Tail-risk regularity.** CVaR on scaled local energy exports a CKN-scale smallness on $1 - \alpha$ mass; this survives aggregation and matches $\mathcal{Q}_{\text{sum}}$.

8.2 Threats to validity (and defenses)

Warning (Adversarial / failure modes). • *Axis cross-talk.* Any adapter that renormalizes across channels could violate $g(r \oplus r') = g(r) \wedge g(r')$. **Defense:** property-based tests in the Adapter suite (golden seeds).

• *Alias drift.* Legacy `omega_trace_id` mismatched with `omega_row_hash`. **Defense:** I5 hard reject; JSON Schema note and harness diff.

• *under-reporting.* A sum-axis emits $\top_{\text{unk}}^{(\beta)}$ but realizes $> \beta$. **Defense:** I4b quarantine is adapter-local and blocks any adaptive move.

• *Frontier under-pricing.* Miscomputed C_{salt} or CI. **Defense:** dimensionless recompute from Digest + row bytes; subadditivity fixture for merged crossings.

8.3 Actionable recommendations (operator playbook)

Checklist (From prototype to practice).

- ☐ **Pin constants in the Digest.** For each run, $\log(N_o, C_B, \nu, C_{\text{CZ}}, C_{\text{LEI}}, N_V, \tau_{\text{min}}, c_{\Sigma}, \kappa_{\text{max}}, r_{\text{tube}}, \hat{\nu}_{\text{min}})$ with hash provenance.
- ☐ **Harden adapters.** Ship monotone-homomorphism tests and an axis-mixing adversary; forbid cross-channel renormalization.
- ☐ **Clamp enforcement.** Byte-list EF “quadratic proxies”; force nonparametric fallback whenever `clamp` $\neq$ Green (I2).
- ☐ **Unknowns policy.** Maintain a signed `beta_registry`; audit realized contributions per sum-axis before any adaptive update (I4b).
- ☐ **Frontier ledger discipline.** Require a per-row CI and a run-level `sis_budg`; fail-closed on overrun (I1).
- ☐ **Reproducibility pack.** Bundle Transport Pack + Reference Digest + container `container_sha` + harness script that recomputes `proof_print`.

8.4 Minimal validation path (de-risking constants)

Tip (Two cheap but decisive dry-runs). (V0) **2D NSE harness.** Freeze LP/CZ as in Part II; run CVaR guard on a smooth 2D solution; require I1–I5 to pass and $\Sigma_{\text{final}} = \emptyset$.

(V1) **Frontier stress test.** Script heavy reassociation within a budget; verify only $\mathcal{Q}_{\text{sum}}$ degrades; check merged-crossing spend $\leq$ rowwise sum; confirm max-channels invariant.

8.5 Limitations

Note (Scope boundaries). The spine assumes availability of an auditable CZ split and a Vitali grid with bounded multiplicity N_V . L5 relies on a classical rigidity statement once ω exports (i)–(iii); strengthening L5 beyond suitable solutions is out of scope here.

8.6 Future directions

1. **Sharper bridge constants.** Replace $C_{\text{CKN}} = C_0 N_V C_{\text{CZ}} C_{\text{LEI}}$ by a locality-aware $C_{\text{CKN}}(r)$ with digest-bound radii, improving slack in I4a.
2. **Adaptive grid pricing.** Integrate data-driven CI estimates for C_{salt} with jackknife/BB to reduce false rejections without weakening I1.
3. **Other PDEs.** Port $(Q, \Sigma, \text{compose_ev})$ to MHD, SQG, or critical NLS where max/sum asymmetry is similarly valuable.
4. **Certified numerics.** Add interval-arithmetic backstops for the RU estimator inside PR to eliminate float nondeterminism in the CVaR guard.
5. **Proof-carrying runs.** Emit a compact, signed `proof_print` transcript per window for third-party verifiers (≤ 1 KB/window).

9 CVaR→CKN: explicit bridge constant and proof sketch

9.1 Objects and notation

For (x_0, t_0) and scale $r > 0$, define the (dimensionful) scaled local energy

$$\mathcal{E}_r[u, p](x_0, t_0) := \frac{1}{r} \int_{t_0-r^2}^{t_0} \int_{B_r(x_0)} \left(|\nabla u|^2 + |p - p_{B_r(x_0)}|^{3/2} \right) dx dt.$$

Let $\text{Grid} = \{(x_k, t_k, r_k)\}_{k \in K}$ be a Vitali grid with bounded multiplicity N_V (digest-bound). For a nonnegative random variable X on a finite measure space, $\text{CVaR}_{1-\alpha}(X)$ denotes the Conditional Value-at-Risk at tail mass $\alpha \in (0, 1)$ (worst α -quantile conditional mean).

Standing constants (Digest bindings (bridge)).

<code>cz_kernel_id</code>	$\mapsto$	C_{CZ} ,	$\mapsto$	C_{LEI} ,	$\mapsto$	<code>nv_multiplicity</code>	$\mapsto$	<code>cvar_bridge_C0</code>	$\mapsto$	C_0 .
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9.2 Bridge constant (operational, digest-bound)

Lemma 9.1 (CVaR→CKN constant). *Fix $\alpha \in (0, 1)$ and a Vitali grid of multiplicity N_V . Assume the PR adapter (Tier-1) computes, for every grid cell k , $\text{CVaR}_{1-\alpha}(\mathcal{E}_{r_k}[u, p]) \leq \varepsilon$. Then there exists a subcover $\mathcal{U} \subset \text{Grid}$ of aggregate measure $\geq 1 - \alpha$ such that, on each cell in $\mathcal{U}$,*

$$\mathcal{E}_{r_k}[u, p] < \varepsilon_{\text{CKN}}, \quad \varepsilon_{\text{CKN}} = C_{\text{CKN}} \varepsilon, \quad C_{\text{CKN}} = C_0 N_V C_{\text{CZ}} C_{\text{LEI}}.$$

The constant C_0 is universal for the bridge step and is fixed to $C_0 = 1$ (declared and bound in the Reference Digest).

Note (Why $C_0 = 1$ is universal). For a nonnegative X , the definition of CVaR implies $\mu\{X > \tau\} > \alpha \Rightarrow \text{CVaR}_{1-\alpha}(X) \geq \tau$. Thus $\text{CVaR}_{1-\alpha}(X) \leq \varepsilon$ forces $\mu\{X > \varepsilon\} \leq \alpha$. No analytic slack is introduced at this tail-to-level step, hence $C_0 = 1$. All additional factors in C_{CKN} come only from CZ/LEI constants and Vitali overlap accounting.

9.3 Proof sketch (auditable chain)

Sketch. Step 1: Tail-to-level smallness (measure-theoretic). For each cell k , apply the CVaR contrapositive cited above to $X = \mathcal{E}_{r_k}[u, p]$: $\text{CVaR}_{1-\alpha}(X) \leq \varepsilon \Rightarrow \mu\{X > \varepsilon\} \leq \alpha$. Selecting $\mathcal{U}$ as the complement of this tail yields a subcollection with measure at least $1 - \alpha$ where $\mathcal{E}_{r_k}[u, p] \leq \varepsilon$.

Step 2: Normalization to the CKN predicate. The classical Caffarelli–Kohn–Nirenberg smallness predicate uses the same local functional up to controlled constants from: (i) the fixed Calderón–Zygmund kernel (pressure split) with operator norm C_{CZ} ; and (ii) the local energy inequality normalization with constant C_{LEI} . These inflate the threshold by $C_{\text{CZ}}C_{\text{LEI}}$.

Step 3: Overlap accounting (Vitali). Passing from per-cell smallness to a subcover carries an at-most- N_V slack due to overlap multiplicity.

Combining Steps 1–3 gives $\varepsilon_{\text{CKN}} = C_0 N_V C_{\text{CZ}} C_{\text{LEI}} \varepsilon$ with $C_0 = 1$. All constituents are Digest-bound and byte-replayable. $\square$

9.4 Bridge replay recipe (Tier-1 harness)

Checklist (Auditor steps).

- ☐ Pull $N_V, C_{\text{CZ}}, C_{\text{LEI}}, C_0$ from the Reference Digest (hash-identified by `cz_kernel_id`, `lei_norm_const`, `nv_multiplicity`, `cvar_bridge_C0`).
- ☐ Verify PR’s per-cell guard: $\text{CVaR}_{1-\alpha}(\mathcal{E}_{r_k}) \leq \varepsilon$ with RU estimator + CI (adapter-local bytes).
- ☐ Compute $C_{\text{CKN}} = C_0 N_V C_{\text{CZ}} C_{\text{LEI}}$ and set $\varepsilon_{\text{CKN}} = C_{\text{CKN}} \varepsilon$.
- ☐ Construct the tail set $\{k : \mathcal{E}_{r_k} > \varepsilon_{\text{CKN}}\}$; assert its measure $\leq \alpha$.
- ☐ Emit `REQUIRESDEPCERT(CVAR)` fail diff if any step does not reproduce.

9.5 Interface to acceptance (I4a)

Ledger (Acceptance hooks).

- **I4a (guard).** The bridge is accepted when the computed ε_{CKN} satisfies the smallness threshold needed by the ε -regularity lemma (L3).
- **I5 (determinism).** All constants used above are declared in the Digest and referenced by IDs; replay must match bitwise under the declared `hash_suite`.

10 Frontier spend: subadditivity lemma and an auditable fixture

10.1 Definitions and units (recall)

A *frontier crossing* c is any non-commuting reassociation (e.g., basis swap, mesh retune) that generates an auditable packet with

$$(dF/\nu, \hat{\nu}_{\min}, c_{\Sigma}, r_{\text{tube}}, \kappa_{\max}, K_{\text{obs}}, \text{CI}) \quad (\text{all bytes present in the row}).$$

The *saltation bound* and the per-row acceptance inequality (dimensionless) are

$$C_{\text{salt}}(c) = \frac{\|F^+ - F^-\|}{\hat{\nu}_{\min}} \left(1 + c_{\Sigma} \kappa_{\max} r_{\text{tube}}\right), \quad K_{\text{obs}}(c) \leq C_{\text{salt}}(c) (1 + \text{CI}(c)).$$

We ledger the *frontier spend* (sum-channel in $\mathcal{Q}$; dimensionless) as

$$\Delta_{\text{fr}}(c) := C_{\text{salt}}(c).$$

(Any alternate but fixed debit rule that is *positively homogeneous and subadditive* w.r.t. $F^+ - F^-$ is equally admissible; we pin the canonical choice above for replayability.)

Note (Digest refinement is monotone). “Refining the Reference Digest” means tightening lower bounds/raising factors: $\hat{\nu}_{\min} \downarrow$, or $c_{\Sigma} \uparrow$, $\kappa_{\max} \uparrow$, $r_{\text{tube}} \uparrow$. Each such change (weakly) *increases* C_{salt} and therefore Δ_{fr} .

10.2 Subadditivity and monotonicity (formal statement)

Lemma 10.1 (Subadditivity and digest monotonicity). *Let $C = \{c_i\}_{i=1}^m$ be a finite multiset of frontier crossings in a window and let $c_{\oplus}$ denote the merged crossing whose geometry difference is the pathwise sum, $F_{\oplus}^+ - F_{\oplus}^- := \sum_{i=1}^m (F_i^+ - F_i^-)$, with the same Digest constants. Then*

$$\Delta_{\text{fr}}(c_{\oplus}) \leq \sum_{i=1}^m \Delta_{\text{fr}}(c_i).$$

Moreover, if $\text{RD} \preceq \text{RD}'$ denotes a refinement of the Digest as above, then

$$\Delta_{\text{fr}}^{\text{RD}}(c) \leq \Delta_{\text{fr}}^{\text{RD}'}(c) \quad \text{for every crossing } c.$$

Sketch. Write $C_{\text{salt}}(c) = \|F^+ - F^-\| A$ with $A := \hat{\nu}_{\min}^{-1} (1 + c_{\Sigma} \kappa_{\max} r_{\text{tube}}) \geq 0$ held fixed by the Digest. Then

$$\Delta_{\text{fr}}(c_{\oplus}) = \|F_{\oplus}^+ - F_{\oplus}^-\| A \leq \sum_{i=1}^m \|F_i^+ - F_i^-\| A = \sum_{i=1}^m \Delta_{\text{fr}}(c_i)$$

by the triangle inequality on the norm $\|\cdot\|$. For monotonicity, note that refinement decreases $\hat{\nu}_{\min}$ and/or increases $c_{\Sigma}, \kappa_{\max}, r_{\text{tube}}$, hence increases A and therefore Δ_{fr} . $\square$

Ledger (Acceptance hooks).

- I1 uses the per-row inequality $K_{\text{obs}} \leq C_{\text{salt}}(1 + \text{CI})$ and the budget $C_{\text{fr}}^+ \leq \text{siss_budget}$.
- I5 fixes the Digest constants and row bytes so that Δ_{fr} is deterministic and replayable.

10.3 Merged-crossing replay (auditor recipe)

Checklist (Deterministic steps).

- ☐ Read each row's $(F_i^+ - F_i^-)$ proxy (the canonical bytes that generate dF/ν) and the common Digest constants $\hat{\nu}_{\min}, c_{\Sigma}, \kappa_{\max}, r_{\text{tube}}$.
- ☐ Compute $C_{\text{salt}}(c_i) = \|F_i^+ - F_i^-\| A$ with A as above; set $\Delta_{\text{fr}}(c_i) = C_{\text{salt}}(c_i)$.
- ☐ Form the merged geometry jump $F_{\oplus}^+ - F_{\oplus}^- := \sum_i (F_i^+ - F_i^-)$ in the same units; compute $\Delta_{\text{fr}}(c_{\oplus}) = \|F_{\oplus}^+ - F_{\oplus}^-\| A$.

- Assert $\Delta_{\text{fr}}(c_{\oplus}) \leq \sum_i \Delta_{\text{fr}}(c_i)$ (subadditivity spot-check).
- Independently verify each per-row acceptance: $K_{\text{obs}}(c_i) \leq C_{\text{salt}}(c_i)(1 + \text{CI}(c_i))$.

10.4 Self-contained fixture (Tier-1; copy/paste for CI)

```

1 # Frontier Spend Fixture (deterministic; dimensionless)
2 digest: { r_tube: 0.50, kappa_max: 1.20, c_sigma: 0.40, nu_min_CI: 0.90 }
3 rows:
4 - frontier_id: fr_a1; op_type: LPRefit; dF_over_nu: 0.30; K_obs: 0.33; CI: 0.02
5 - frontier_id: fr_b7; op_type: PatchRelabel; dF_over_nu: 0.12; K_obs: 0.10; CI: 0.02
6 - frontier_id: fr_c3; op_type: MeshRetune; dF_over_nu: 0.22; K_obs: 0.18; CI: 0.02
7
8 saltation_formula:
9   C_salt = dF_over_nu * (1 + c_sigma*kappa_max*r_tube) / nu_min_CI
10
11 computed:
12   A = (1 + 0.40*1.20*0.50) / 0.90 = 1.266666...
13   C_salt: { a1: 0.30*A = 0.380, b7: 0.12*A = 0.152, c3: 0.22*A = 0.279 }
14   per-row checks: K_obs <= C_salt*(1+CI) holds with CI=0.02
15   sum_spend = 0.380 + 0.152 + 0.279 = 0.811
16   merged_crossing:
17     dF_over_nu_merged = 0.30 + 0.12 + 0.22 = 0.64
18     C_salt_merged = 0.64*A = 0.811 <= sum_spend # subadditivity (tight here)

```

Tip (Operator guidance). Budget `sisss_budget` against the *sum* of row debits; the acceptance inequality is checked per-row, while subadditivity ensures merged operations never hide extra spend. Because Δ_{fr} is dimensionless, it composes cleanly across windows and Digest versions (monotone under refinement).

11 Determinism envelope: compliance harness (auditor-side)

Note (Goal). Given a *Transport Pack* and *Reference Digest*, the harness replays all Tier-1/2 computations *bitwise* under a fixed `hash_suite` and container image, emits a PASS/FAIL verdict, and prints a minimal `diff` for any reject. All checks here implement `I5` and call out

`I1` / `I4a` / `I4b` where relevant.

11.1 Envelope invariants (must hold verbatim)

Checklist.

- **Suite lock.** `hash_suite = {name, version}` is constant within a run; mixed suites $\Rightarrow$ `I5` reject.
- **Canonical serialization.** All hashes use *canonical JSON* (UTF-8; sorted keys; integers in base-10 without leading zeros; floats as IEEE-754 `float64` roundTiesToEven marshaled via decimal without NaN/Inf; no trailing zeros policy enforced by schema). The harness exposes

- a single function `stable_bytes(·)`.
- ☐ **Alias rule.** If `omega_trace_id` is present, it must equal `omega_row_hash` byte-for-byte; else I5 reject.
- ☐ **Determinism envelope complete.** Required fields: `{seed, clock_ms, ntp_offset_ms, monotone_counter, float_format}` = IEEE-754 float64, tolerances = {abs, rel}, container_sha}.
- ☐ **Clock discipline.** `clock_ms` nondecreasing across windows; $|ntp_offset_ms| \leq$ declared bound; `monotone_counter` strictly increases (no wraparound).
- ☐ **Numeric mode.** Denormals preserved (no FTZ/DAZ), rounding mode `to-nearest-even`; the harness probes these at runtime inside the container and fails if mismatched.

☐ Determinism	envelope	complete.	Required	fields:
<code>{seed, clock_ms, ntp_offset_ms, monotone_counter, float_format</code>				=
<code>IEEE-754 float64, tolerances = {abs, rel}, container_sha}.</code>				

□ **Numeric mode.** Denormals preserved (no FTZ/DAZ), rounding mode `to-nearest-even`; the harness probes these at runtime inside the container and fails if mismatched.

11.2 Replay procedure (single window)

```

1 # Determinism Envelope Harness (auditor-side, single window)
2 inputs:
3   transport_pack: {...}      # canonical JSON object (window-scoped)
4   ref_digest: {...}         # canonical JSON object (run-scoped)
5 procedure:
6   1. Schema & suite:
7     - Validate required keys/types (V1). Fail  $\rightarrow$  {stage: V1, item: schema, diff: ...}
8     - Assert hash_suite is run-consistent (I5).
9   2. Container boot:
10    - Launch container_sha; set RNG seed=seed; pin float mode; export tolerances.
11    - Print runtime probe of rounding/denormals/BLAS threading for the record.
12   3. Canonicalization:
13    - B := stable_bytes(transport_pack.subset_for_proof_print)
14    - Recompute proof_print := H(B) under hash_suite. Compare hex  $\rightarrow$  else {I5, diff}.
15   4. Frontier rows (I1):
16    - For each row: read (dF/nu, nu_min_CI, c_sigma, r_tube, kappa_max, CI, K_obs).
17    - Compute C_salt = (dF/nu) * (1 + c_sigma*kappa_max*r_tube) / nu_min_CI.
18    - Assert K_obs <= C_salt*(1+CI). Budget: sum(C_salt) <= siss_budget.
19    - Spot-check subadditivity via merged-crossing recompute.
20   5. Clamp/proxy policy (I2):
21    - If clamp in {Amber, Red} then auto-emit RetuneDisabledInRed; verify PR  $\leftarrow$ 
22       $\hookrightarrow$  discharge only if EF turned Green same window.
23    - Reject on proxy use when clamp  $\neq$  Green.
24   6. DRO sensitivity (I3):
25    - Check thresholds on (dhat_rho, dualgap); quarantine on breach (reject).
26   7. PR CVaR guard (I4a):
27    - RU estimator replay with seed; compute CVaR_{1-alpha}(E_r) per cell.
28    - Pull (N_V, C_CZ, C_LEI, CO) from Digest; set eps_CKN = CO*N_V*C_CZ*C_LEI*eps.
29    - Verify measure{E_r > eps_CKN} <= alpha. Else {I4a, diff}.
30   8. Unknowns cap (I4b):
31    - For each sum-axis with top_unk(beta): compute realized contrib_sum <= beta.
32    - On breach: set quarantine=1; reject before any adaptive updates.
33 outputs:
34   verdict: PASS | FAIL
35   diffs: [{path: "...", expected: ..., got: ..., tol: {abs, rel}}]

```

11.3 Diff surface (minimal and stable)

Field	Type	Expected	Got	On fail	
proof_print	hex	$H(\text{stable_bytes})$	hex	I5	reject
omega_row_hash	hex	equals omega_trace_id (if present)	hex	I5	reject
C_salt (row i)	real	recompute	value	I1	reject
K_obs (row i)	real	$\leq C_{\text{salt}}(1+CI)$	value	I1	reject
sisss_budget	real	$\geq \sum C_{\text{salt}}$	value	I1	reject
CVaR cell k	real	$\leq \varepsilon$	value	I4a	reject
ε_{CKN}	real	$C_0 N_V C_{CZ} C_{\text{LEI}} \varepsilon$	value	I4a	reject
contrib_sum[a]	real	$\leq \beta$	value	I4b	reject

11.4 Negative controls (must deterministically fail)

- Warning (Harness-internal tests).** ☐ Remove a REQUIRED key $\Rightarrow$ V1 fail with path of missing field.
- ☐ Mix hash_suite versions within a run $\Rightarrow$ I5 fail with first offending window id.
- ☐ Flip rounding mode or enable FTZ/DAZ $\Rightarrow$ I5 fail (numeric mode mismatch).
- ☐ Proxy enabled while clamp=Red $\Rightarrow$ I2 fail.
- ☐ $\mathbb{T}_{\text{unk}}^{(\beta)}$ realizes $> \beta \Rightarrow$ quarantine+reject (I4b).
- ☐ CVaR RU replay with different seed changes cell decisions $\Rightarrow$ I5 fail (seed noncompliance).
- ☐ omega_row_hash $\neq$ omega_trace_id (when alias present) $\Rightarrow$ I5 fail.

11.5 Harness configuration (portable defaults)

```

1 # harness_config.yml (portable defaults)
2 float:
3   format: "IEEE-754 float64"
4   rounding: "to-nearest-even"
5   tolerances: { abs: 1e-12, rel: 1e-9 }
6   denormals: "preserve" # reject if FTZ/DAZ detected
7 random:
8   seed_source: "transport_pack.determinism_envelope.seed"
9 container:
10  sha: "<container_sha>"
11  threads: 1 # single-thread BLAS/LAPACK for bitwise stability
12 hashing:
13  suite: { name: "BLAKE3-256", version: "H_v1" }
14 serialization:
15  mode: "canonical-json" # stable_bytes() applies RFC-8785-style rules

```

11.6 Where I5 meets other items

Ledger (Cross-links).

- I1 : The frontier inequality is recomputed inside the container; failures are *deterministic* under the envelope.
- I4a : RU-based CVaR estimates are seeded and serialized; the bridge constant is pulled from the Digest and hashed.
- I4b : Unknowns policy is adapter-local; enforcement happens before any adaptive update and is logged in the diff.

12 Adapter contract: property-based tests (golden seeds)

Note (Test intent). Adapters $g : \text{Raw} \rightarrow \mathcal{Q}$ must be *monotone homomorphisms*: $g(r \oplus r') = g(r) \wedge g(r')$ and $r \succeq r' \Rightarrow g(r) \succeq g(r')$. The seeds below are minimal, reproducible fixtures that an auditor can run verbatim.

12.1 Monotone homomorphism tests (golden)

```
1 # Seed: seed_mh_001 (ASCII-only)
2 # Goal: check  $g(r \oplus r') == g(r) \wedge g(r')$  and isotonicity in the Raw preorder.
3 gen:
4   # Draw independent Raw payloads; mask in {dropout, clipped(beta), nominal}.
5   r1, r2 = RawGen.sample_pair(seed="seed_mh_001")
6   assert RawPreorder.is_total_on_required_axes()
7
8 ops:
9   r_plus = RawMonoid.oplus(r1, r2)      # concatenation/merge w/o renorm
10  g_plus = g(r_plus)
11  g_meet = meet(g(r1), g(r2))           # coordinatewise: max on Q_max, sum on Q_sum
12
13 assertions:
14  assert g_plus == g_meet
15  if RawPreorder.ge(r1, r2):             # r1 >= r2 (worse-is-greater convention)
16    assert QPreorder.ge(g(r1), g(r2))
```

12.2 Adversarial nonexample: renormalizing monoid (should fail)

```
1 # Seed: seed_nonexample_renorm (ASCII-only)
2 # Goal: any cross-stream renormalization must violate the homomorphism law.
3 gen:
4   r1, r2 = RawGen.sample_pair(seed="seed_nonexample_renorm")
5
6 define_bad_monoid:
7   def opus_bad(a, b):
8     payload = concat(a.payload, b.payload)
9     return renorm_unit_variance_per_stream(payload) # ILLEGAL: changes units/scale
10
11 ops:
```

```

12  r_bad   = oplus_bad(r1, r2)
13  g_bad   = g(r_bad)
14  g_meet  = meet(g(r1), g(r2))
15
16  assertions:
17  assert g_bad  $\neq$  g_meet           # must fail; CI should reject this implementation

```

12.3 Positive clip-merge on a sum-axis (policy carries through)

```

1  # Seed: seed_positive_clip_merge (ASCII-only)
2  # Goal: clipped stream contributes at most beta_cap to the sum-axis; law preserved.
3  policy:
4    beta_cap = 0.20
5    axis = "tail_mass_sum"           # example sum-channel name
6
7  gen:
8    r_nom    = RawGen.nominal(seed="seed_positive_clip_merge_nom")
9    r_clip   = RawGen.clipped(axis=axis, beta=beta_cap,  $\hookleftarrow$ 
10                $\hookrightarrow$  seed="seed_positive_clip_merge_clip")
11
12  ops:
13    r_plus   = RawMonoid.oplus(r_nom, r_clip)
14    g_nom    = g(r_nom)
15    g_clip   = g(r_clip)
16    g_plus   = g(r_plus)
17    g_meet   = meet(g_nom, g_clip)
18
19  assertions:
20  assert contrib_sum(g_clip, axis) <= beta_cap
21  assert g_plus == g_meet

```

12.4 Unknowns budget breach (14b) triggers quarantine

```

1  # Seed: seed_unk_012 (ASCII-only)
2  # Goal: if realized contribution exceeds beta_cap on a sum-axis, quarantine must fire  $\hookleftarrow$ 
3            $\hookrightarrow$  and adaptive updates block.
4  policy:
5    beta_cap = 0.10
6    axis = "tail_mass_sum"
7
8  gen:
9    r_unk = RawGen.unknowns(axis=axis, beta=beta_cap, seed="seed_unk_012")
10    # Force realized contribution above cap in this synthetic case:
11    realize_contrib_sum(r_unk, axis, value=0.13)
12
13  ops:
14    q = g(r_unk)
15
16  assertions:
17  assert q.flags.quarantine == 1

```

```

17  assert q.flags.reason == "UNK_BUDGET_EXCEEDED"
18  assert q.flags.adaptive_updates_blocked is True
19  assert q.flags.monitors_continue is True

```

12.5 Axis-mixing adversary (guards against cross-channel renorm)

```

1  # Seed: seed_axis_mix_001 (ASCII-only)
2  # Goal: prevent an adapter from renormalizing a max-channel using a sum-channel (or  $\leftrightarrow$ 
    $\hookrightarrow$  vice versa).
3  gen:
4      r_energy_simple = RawGen.energy_only(seed="seed_axis_mix_E")
5      r_cvar_complex = RawGen.cvar_only(seed="seed_axis_mix_C")
6
7  ops:
8      r_plus = RawMonoid.oplus(r_energy_simple, r_cvar_complex)
9      g_E = g(r_energy_simple)
10     g_C = g(r_cvar_complex)
11     g_plus = g(r_plus)
12     g_meet = meet(g_E, g_C)
13
14  assertions:
15     assert g_plus == g_meet
16     # Additional invariant: changing a sum-axis must not alter any max-axis coordinate.
17     assert QMax.project(g_plus) == QMax.project(g_meet)

```

Tip (What the auditor observes). Each seed fixes the generator, so a third party can reproduce the same Raw payloads and byte-for-byte adapter outputs. Failures are localized: the harness prints a single diff showing the violated identity or bound, tagged with the seed.

13 Acceptance shocks: how rapid changes in acceptance produce humor, disproportional radicalism, and strong emotions

Concept (Shock state and metric (operational)). Let the *acceptance state* at time t be

$$A_t := (\text{clamp}_t \in \{\text{Green} \preceq \text{Amber} \preceq \text{Red}\}, \quad \Sigma_t \text{ (multiset of obligations)}, \quad Q_t = Q_{\max,t} \wedge Q_{\text{sum},t}).$$

Fix a window-scale duration τ_{shock} (digest-bound) and weights $\lambda_{\text{clamp}}, \lambda_{\Sigma} > 0$, and a diagonal $W \succeq 0$ on Q_{sum} (all *dimensionless*). Define the *shock seminorm* for adjacent event times $t^- < t^+$ within τ_{shock} :

$$\|A_{t^+} - A_{t^-}\|_* := \lambda_{\text{clamp}} \cdot \mathbf{1}\{\text{clamp jumps}\} + \lambda_{\Sigma} \cdot |\mu(\Sigma_{t^+}) - \mu(\Sigma_{t^-})| + \|W \cdot (Q_{\text{sum},t^+} - Q_{\text{sum},t^-})\|_1.$$

An *acceptance shock* occurs if $\|A_{t^+} - A_{t^-}\|_* \geq \theta_{\text{shock}}$ for some threshold θ_{shock} (digest-bound).

Rule (Sum-only debit preserves max-gates; derivatives drive affect). By design, frontier spend debits *only* Q_{sum} , while $Q_{\max}$ (e.g., $E, S_3, P_{H-1}, Q_{\tau}$) is reassociation-invariant under `compose_ev`. Thus max-gates do not change under reassociation, but *human perception* tracks $\partial_t Q_{\text{sum}}$ and sudden emissions into Σ , so large t -derivatives amplify affect even when gate truths are invariant.

Three archetypes (diagnostics + mitigations). **(H) Humor: benign violation, fast repair.** A small Green→Amber blip (e.g., I4b cap trip or missing DEPCERT) immediately rectified in-window. The narrative is violation→replay→deterministic fix; Q_{sum} shows a brief hump; Σ descends to baseline.

Note (Audit cues for (H)). Short dwell in Amber; $\Delta\mu(\Sigma) \downarrow$ to zero within one window; Σ -debit subadditive and small; PR witness present; per-row frontier checks clean.

(R) Radicalism: Red lock induces overshoot. A fast Amber→Red jump (e.g., No-Zeno breach with exchange-slack fail) auto-emits FRONTIERFREEZE, RETUNEDISABLEDINRED. Attempted “clever” adaptive moves under Red must hard-fail I2, creating social pressure for outsized refactors.

Warning (Mitigation for (R)). Pin proxies-only-when-Green (I2). Provide a monotone, nonparametric fallback under Amber/Red so motion is still *possible* without rule-breaking. Display a bounded-response counter K to reinforce that the system remains under control.

(E) Emotion: tail-risk surprise. When $\text{CVaR}_{1-\alpha}$ has long run stability and then fails after a frontier burst, the perceived surprise is the gap between expected and realized tail mass, not merely the binary pass/fail.

Tip (Mitigation for (E)). Publish forecast CVaR with a CI band and print the exact cells that moved from body to tail on failure. Turn opaque tail shifts into explicit diffs; this dampens affect.

Replay recipe (shock ledger slice)

1. Identify adjacent event times $t^- < t^+$ with $t^+ - t^- \leq \tau_{\text{shock}}$ (from envelope clocks).
2. Compute Δclamp , $\Delta \mu(\Sigma)$, and $\Delta \mathcal{Q}_{\text{sum}}$; evaluate $\|A_{t^+} - A_{t^-}\|_*$.
3. If Green $\leftrightarrow$ Red without intermediate Amber, print a *shock row* requiring a post-mortem ledger slice (AS1).
4. Tag the archetype: (H) if $\mu \downarrow 0$ in-window and spend small; (R) if Red lock persists; (E) if tail mass jumped and CVaR failed.

Acceptance checks.

1. **AS1 (Clamp volatility).** Flag any Green $\leftrightarrow$ Red jump without Amber; require a one-window post-mortem ledger slice and narrative.
2. **AS2 (Unknowns transparency).** If any sum-axis uses $\top_{\text{unk}}^{(\beta)}$, print realized contribution and registry provenance inline; if value $> \beta$, set **quarantine=1** and *reject before adapt*.
3. **AS3 (Narrative replay).** For every shock, print which predicate flipped, the byte witness, and which discharge closed it; no story $\Rightarrow$ reject.
4. **AS4 (Tail shock hygiene).** On CVaR failure, print $(\alpha, \varepsilon, \varepsilon_{\text{CKN}})$, the count/measure of failing cells, and the diff set; require bridge recompute (I4a).
5. **AS5 (Red lock guard).** Under **clamp=Red**, enforce **RETUNEDISABLEDINRED** and **FRONTIERFREEZE**; any adaptive attempt $\Rightarrow$ I2 hard fail.

Example 1 (Two-window humor arc (H)). Window t : CVaR guard fails at $\alpha = 0.1$, PR emits **REQUIRESDEPCERT(CVaR)**; **clamp=Amber**. Window $t+1$: same grid recomputed; mis-bucketed cells corrected; $\text{CVaR}_{0.9}$ passes; discharge occurs; ledger spend small and subadditive. Reviewers experience a violation $\rightarrow$ repair arc—"the joke lands."

Example 2 (Red lock & overshoot (R)). A No-Zeno breach plus exchange-slack fail bumps **clamp** to Red; **FRONTIERFREEZE** and **RETUNEDISABLEDINRED** auto-emit. Operators propose a retune; I2 blocks it (byte-checked). Provide a nonparametric fallback that restores Green in two clean windows; radical calls for a rewrite subside.

Example 3 (Tail surprise (E)). After many clean windows with stable $\text{CVaR}_{0.95}$, a frontier-heavy reassociation shifts a few cells over the CKN threshold. The bridge recompute prints $\varepsilon_{\text{CKN}} = C_{\text{CKN}}\varepsilon$, lists the offending cells, and shows that max-gates (e.g., S_3) are unchanged. Affect is tempered by transparent diffs.